Find the mean marks in statistics and the coefficient of correlation between marks in the two subjects when the variance of Y is 25.

Solution:

We are given that:

$n = 50$, Regression equation of X on Y as $3Y - 5X + 180 = 0$

$\bar{y} = 50$, $V(X) = \frac{16}{25} V(Y)$, and $V(Y) = 25$.

We have to find (i) $\bar{x}$ and (ii) r_{XY}

(i) Calculation for $\bar{x}$

Since $(\bar{x}, \bar{y})$ is the point of intersection of the two regression lines, they lie on the regression line $3Y - 5X + 180 = 0$

Hence, $3\bar{y} - 5\bar{x} + 180 = 0$

$3(50) - 5\bar{x} + 180 = 0$

$-5\bar{x} = -180 - 150$

$= -330$

$\bar{x} = \frac{-330}{-5} = 66$

$\bar{x} = 66$

(ii) Calculation for coefficient of correlation.

$3Y - 5X + 180 = 0$

$-5X = -180 - 3Y$

$X = 36 + 0.6\ Y$

$b_{XY} = 0.6$

Also $b_{XY} = r_{XY} \frac{SD(X)}{SD(Y)}$

$0.6 = r_{XY} \frac{SD(X)}{SD(Y)}$

$r_{XY} = \frac{0.6 \times SD(Y)}{SD(X)}$

$r_{XY}^2 = 0.36 \times \frac{V(Y)}{V(X)}$ (1)

Given that:

$V(Y) = 25$

$V(X) = \frac{16}{25} V(Y)$

$= \frac{16}{25} \times 25$

$V(X) = 16$

12th_Statistics_EM_Unit_5.indd 145 1/28/2020 6:36:48 PM